

VEDAT ZEYBEK

Software Engineering Student | Backend Engineering & AI Systems

Turkey | vedatzeybek20@gmail.com | +90 545 476 21 36 | GitHub | LinkedIn

SUMMARY

Fourth-year Software Engineering student with experience in backend development, AI systems, and systems programming. Hands-on with Java, Python, Spring Boot, PostgreSQL, Docker, Linux, REST APIs, C, and C++.

EXPERIENCE

ASELSAN - Software Engineering Intern

Jul 2026 - Sep 2026

- Developed a persistent knowledge and memory system for AI-assisted software engineering workflows.
- Built backend services using Java, Spring Boot, PostgreSQL, pgvector, and Docker, including search and LLM integrations.
- Improved benchmark pass rate from 65.1% to 73.0% with persistent engineering memory and hybrid retrieval.

Orion Innovation - Software Engineering Intern

Sep 2025 - Jun 2026

- Contributed to a variety of software development tasks across backend services, internal tools, integrations, and AI-assisted workflows.
- Worked with different codebases and project requirements, helping implement features, troubleshoot issues, and improve existing components. This included contributing to a Nokia project as part of the internship.
- Collaborated in day-to-day development and maintenance workflows using APIs, Docker, Git, and internal engineering tools.

SELECTED PROJECTS

VitrA Press AI Assistant - 1st Place, Eczacibasi EnGenius Project Competition

- Built an AI-powered troubleshooting assistant for VitrA industrial press operations, using RAG and technical documentation to deliver structured diagnostic and maintenance guidance.

Community Management System - Backend / Collaborative Software Project

- Contributed to a multi-service community platform, implementing and debugging backend workflows, APIs, authorization logic, and database-backed features.

Webserv - Non-Blocking HTTP/1.1 Server | C++98

- Built an event-driven HTTP/1.1 server using sockets and epoll, supporting routing, CGI, uploads, redirects, keep-alive, and timeouts.

Minishell - Unix Shell | C

- Implemented a Unix shell with command parsing, environment variables, built-ins, external commands, pipes, and redirections.

Brand Logo Detection - Computer Vision | Python

- Trained and compared object-detection models including YOLOv7 and evaluated performance using mAP, precision, and recall.

EDUCATION

Bahcesehir University - B.Sc. Software Engineering

2022 - Expected Jul 2027

GPA: 3.40 / 4.00

42 Istanbul - Common Core

2024 - Present

Project-based software engineering training in C, algorithms, Unix/Linux tooling, team projects, and code reviews.

TECHNICAL SKILLS

Languages: Python, Java, C++, C

Backend & Data: Spring Boot, FastAPI, REST APIs, PostgreSQL, pgvector, Redis

AI Systems: LLM APIs, Embeddings, RAG, MCP, Agentic Systems

Infrastructure: Docker, Docker Compose, Linux, Prometheus, Grafana, Git, GitHub

LANGUAGES

Turkish: Native | **English:** B2 | **German:** A2